



Crowd Evacuation

*Simulating the Dynamics
of a Mass Event Disaster*



Distributed AI @ UNIMORE
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Why



Imagine being in a crowded public square and suddenly, something goes wrong. The reason could be many things: a panic stampede, a fire, a structural collapse, or simply a false alarm.

Whatever the trigger, the crowd **must evacuate, fast.**

In these scenarios, the **real danger emerges from the crowd itself**: bottlenecks at exits, herding behaviour, slower individuals being overwhelmed, and panic spreading faster than information.

The denser the crowd, the more these dynamics amplify injury and mortality.

This project addresses exactly this problem.

I built an **Agent-Based Model in NetLogo** that **simulates the evacuation of a crowd** of people.

The model also includes optional **hazards**, that allowed me to study how environmental dangers interact with crowd behaviour under stress.



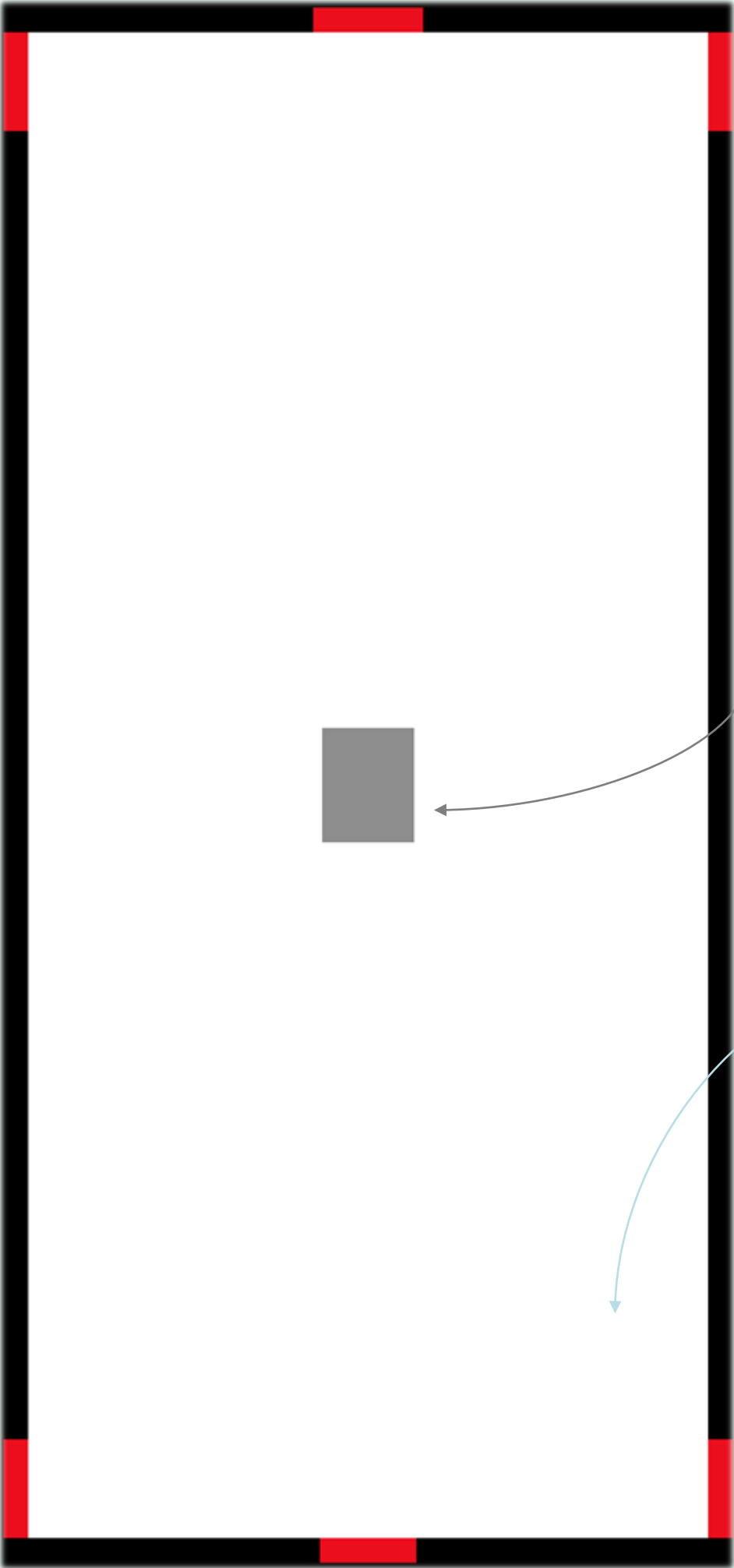
Arena

Walls

Obstacle

Walkable surface

Exits



Arena

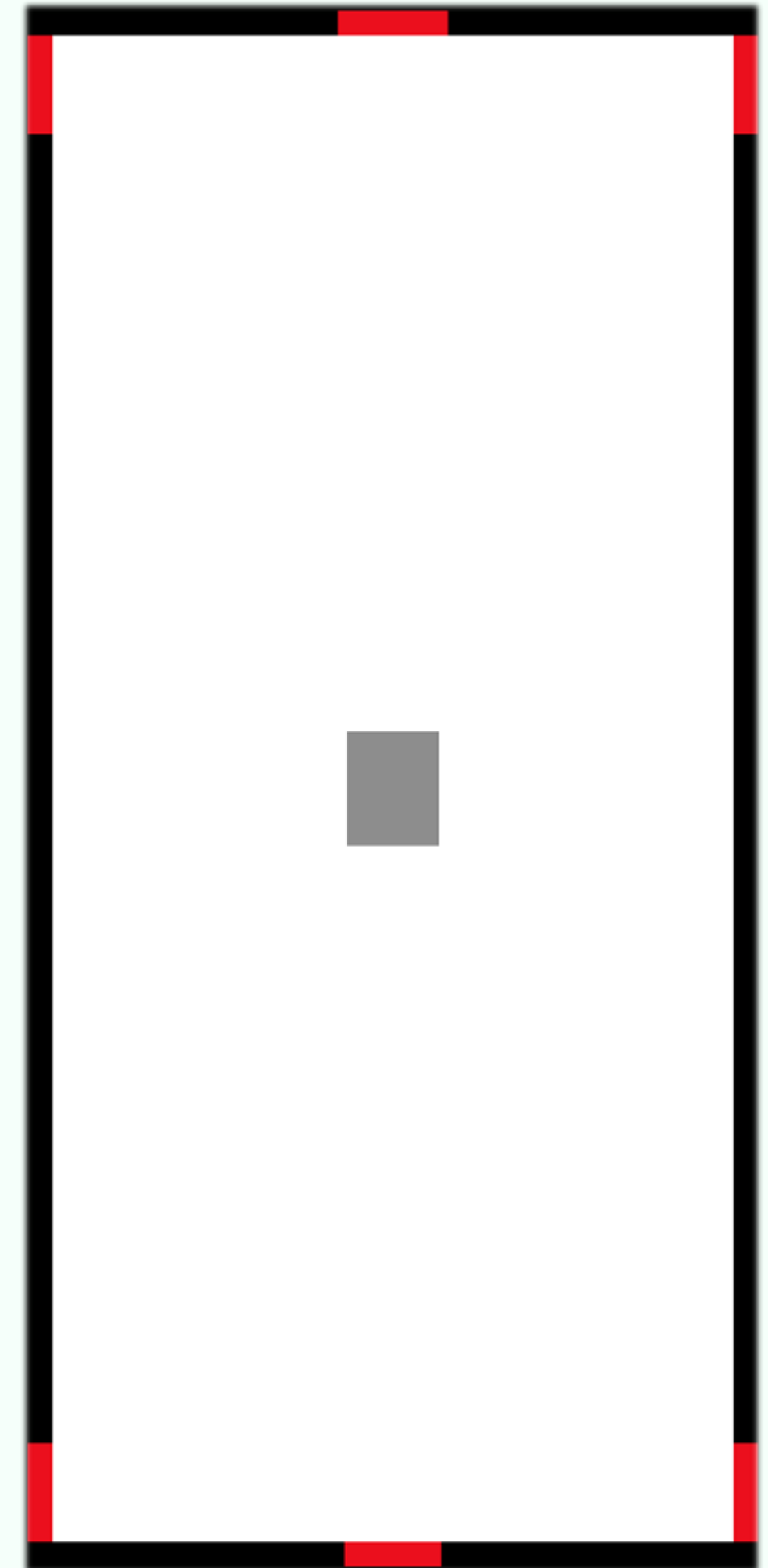
The structure of the arena can be controlled dinamically with the following parameters:

- *scale* change the scale of the arena
- *wall-thickness* change the thickness of the walls
- *real_exits*
 - If false → 6 exits
 - If true → 5 exits

scale

wall-thickness

real_exits

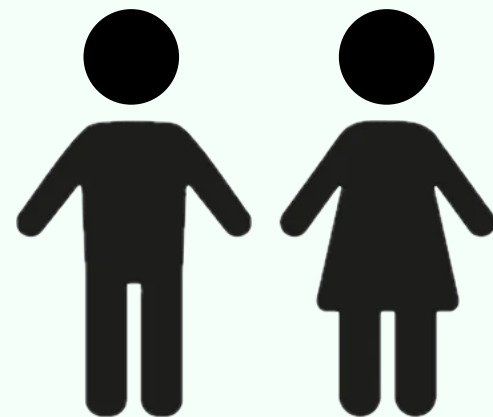
☒

Population

The number of **people** present in the setup can be chosen with a slider and goes from 1 to 30.000

The population can be differentiated along two demographic dimensions:

Gender



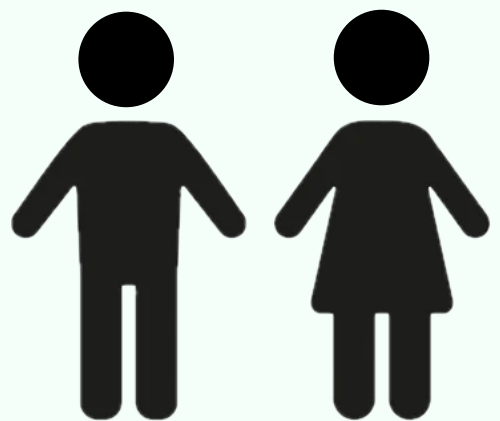
Age



These attributes are assigned randomly to subsets of the population according to configurable sliders in the interface

Population

Gender



Gender affects movement speed

	<i>Male</i>	<i>Female</i>
Speed	Standard	1 step per tick slower

Population

Age

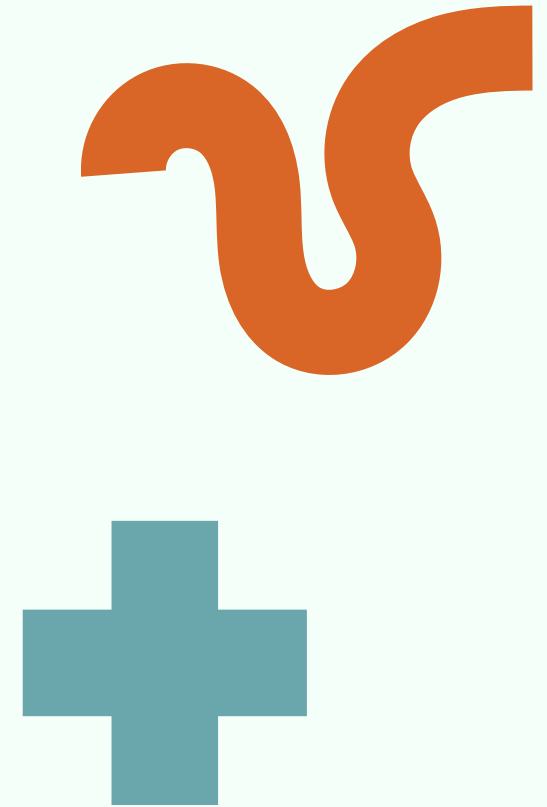


Age affects movement speed

	<i>Children</i>	<i>Adult</i>	<i>Elderly</i>
Speed	2 step per tick slower	Standard	3 step per tick slower

Population

Age



Age acts as a multiplier on crowding damage.

Crowding damage is the health loss that an agent suffers each tick when sharing a patch with other people.

This models the physical harm caused by being pressed, pushed, or trampled in a dense crowd.

Population

Age



Age acts as a multiplier on crowding damage.

	<i>Children</i>	<i>Adult</i>	<i>Elderly</i>
Crowding damage	× 3	× 1	× 2

Children and elderly people lose health at a much faster rate than adults when caught in a dense crowd.

Population

All these penalties are cumulative...

Example:

Penalty given by the gender = speed-1
Penalty given by the age = speed-3

Result = speed-4

She also takes double the crowding damage of an adult male in the same conditions.

As a result, **children and elderly agents are the most vulnerable groups** in the simulation



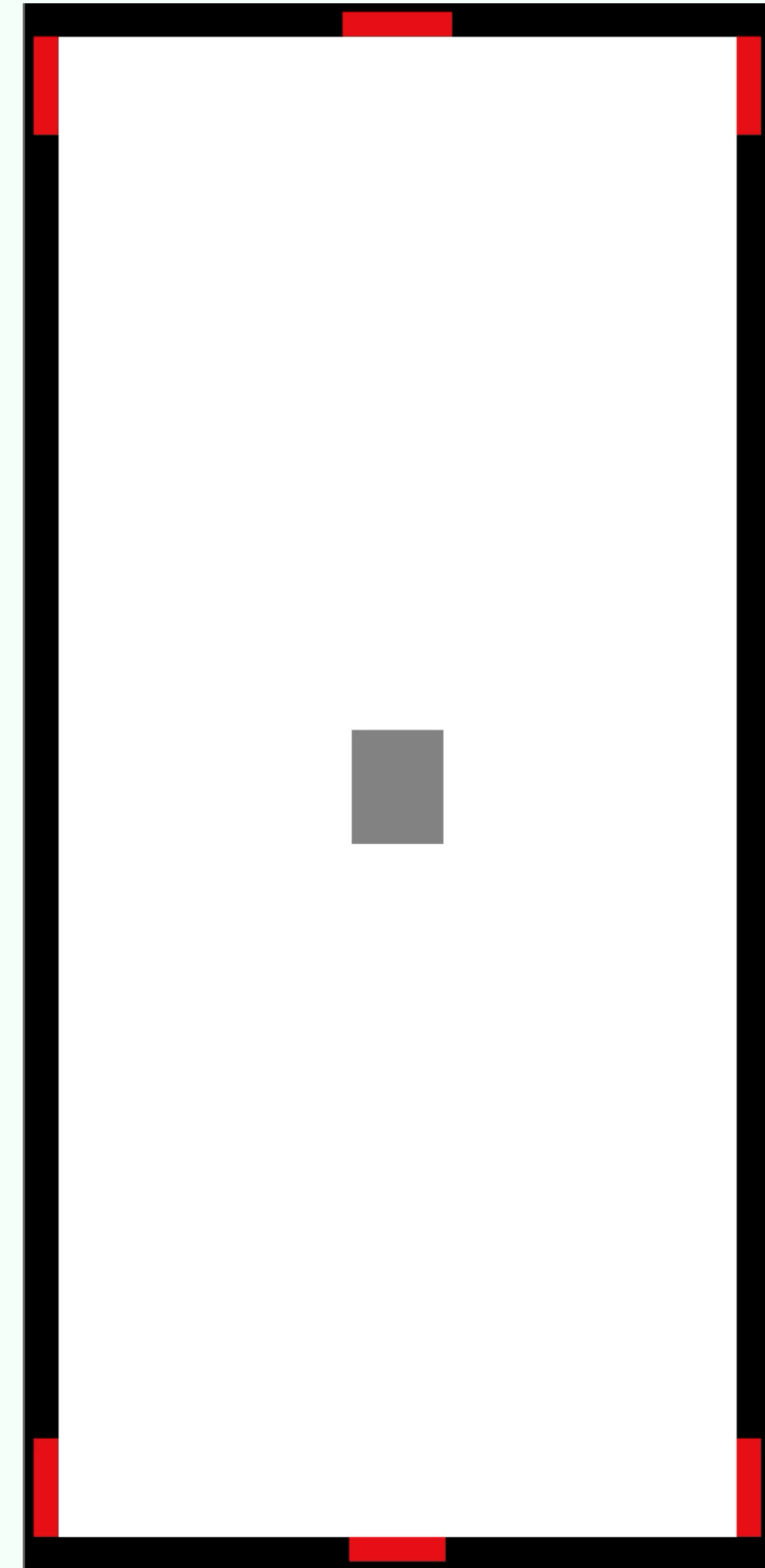
Speed can be set at 1
step per tick for
everyone if
`speed_enabled=false`

Agent Behaviour

Pre Alarm Wandering

Before the evacuation alarm, agents move freely using a **weighted random walk**. At each tick, they randomly turn (left or right), adjust their direction, and move forward if the next patch is walkable.

This phase models natural, unguided crowd movement, with no specific goal or coordination.



Agent Behaviour

Alarm and Destination Assignment

When the alarm is triggered, all agents switch to evacuation mode and are assigned an exit.

Aware agents identify the **nearest gate** to their current position and select a random patch within that gate as their destination.

While **unaware agents** are assigned a completely **random** patch from any gate, regardless of distance.

Once a destination is set, agents orient toward it and begin evacuating.

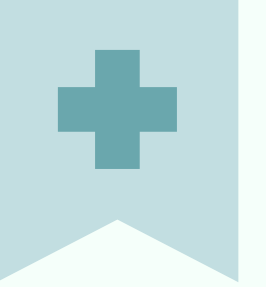


Agent Behaviour

Evacuation Movement – Standard Behaviour

Each tick, agents move up to a fixed number of steps, following a priority-based logic:

- **Fire ahead:** immediately avoid it by moving to the safest neighboring patch; otherwise stay still.
 - **Free floor:** move forward.
 - **Crowded floor:** detour through a less congested neighboring patch.
 - **Free exit:** pass through and evacuate.
 - **Crowded exit:** reroute to a less crowded gate and move sideways.
 - **Wall:** try a nearby exit, otherwise slide along it or pick a new gate.
 - **Statue obstacle:** choose a new random destination and proceed if possible.



Agent Behaviour

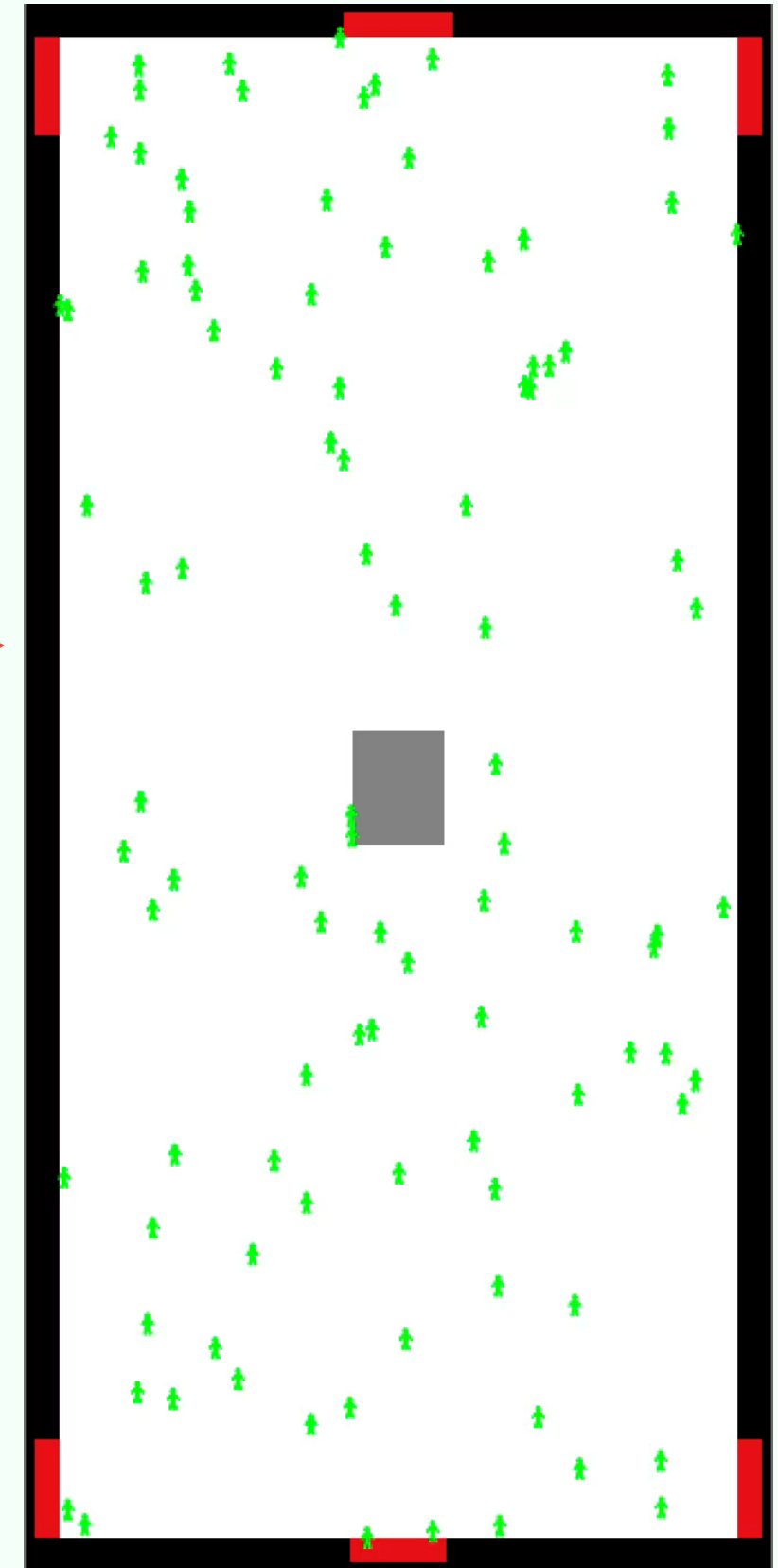
Evacuation Movement – Panic Behaviour

Panicking agents do not always follow the standard movement logic.

At each tick, they decide whether to act rationally or irrationally based on their *panic percentage*:

- **Rational behavior:** follow the standard movement logic.
- **Irrational behavior:** (*follow_crowd*) moving toward the most crowded neighboring patch → herding effect.

If an exit is adjacent, they evacuate regardless of panic.



The injury scale



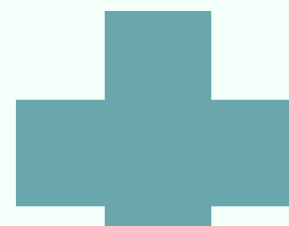
Agent health is modelled through a continuous variable *health_state*, which starts at 100 for every agent and decreases over time as a result of the hazards encountered during the evacuation.

From this value, a **discrete injury level** is derived, ranging from 0 to 6 and directly inspired by the *Abbreviated Injury Scale (AIS)*, a standardised medical classification of trauma severity.

- 0 Healthy
- 1 Minor
- 2 Moderate
- 3 Serious
- 4 Severe
- 5 Critical
- 6 Fatal

The progression of injury levels across the population is tracked in real time through **dedicated counters**

Healthy 618	Evacuated 980
Minor 304	Moderate 48
Serious 8	Severe 1
Critical 1	Fatal 20



The injury scale

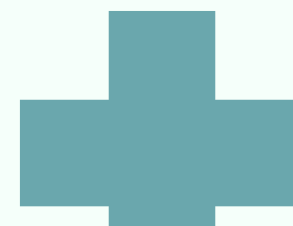
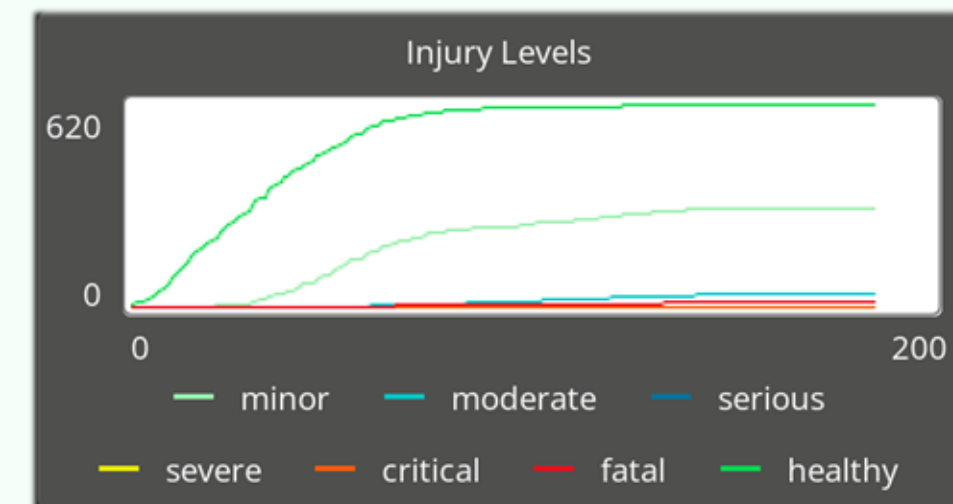


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The progression of injury levels across the population is tracked in real time through a **histogram plot**, which is updated each time an agent successfully evacuates, giving a clear picture of the overall damage sustained by the crowd.



Sources of damage

Crowding Damage and Behavioural Factors

The primary source of harm in the simulation is **crowding**.

Each tick, an agent loses health proportionally to the number of other alive agents standing on the same patch, scaled by a global *injury_weight* parameter.

Two behavioural parameters directly influence how congestion forms and where it concentrates.

The *panic_fraction* determines the **share of agents who, during the evacuation, periodically lose rational decision-making** and instead follow the crowd (they move toward the most densely occupied neighbouring patch).

This **herding behaviour** tends to create **dangerous bottlenecks**, as panicking agents cluster together rather than distributing themselves across available exits, dramatically increasing local densities and therefore crowding damage.

See slide 14: Agent Behaviour – Evacuation Movement – Panic Behaviour

The *aware_fraction*, on the other hand, controls the **share of agents who know the arena layout and can identify the nearest exit**.

While awareness generally **reduces evacuation time**, it can paradoxically **increase injuries** when the fraction is very high, because a large proportion of agents converge on the same nearest exit simultaneously, creating severe congestion near the gates and elevating crowding damage precisely in the most critical area of the evacuation.

See slide 27: Experiments – Aware Agents



Sources of damage

Glass Bottles

When the *glass_bottles* switch is enabled, the ground is treated as **partially covered with broken glass**, introducing a **slipping hazard**.

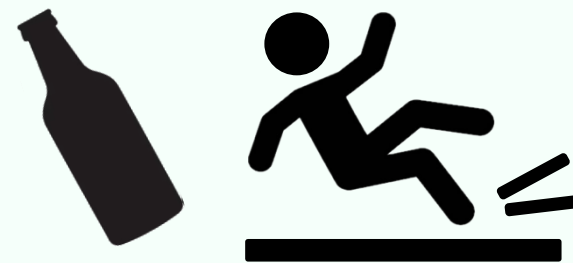
glass_bottles

At each movement step, every agent samples a Bernoulli variable with probability *slipping_chance* / 100.

slipping_chance

3

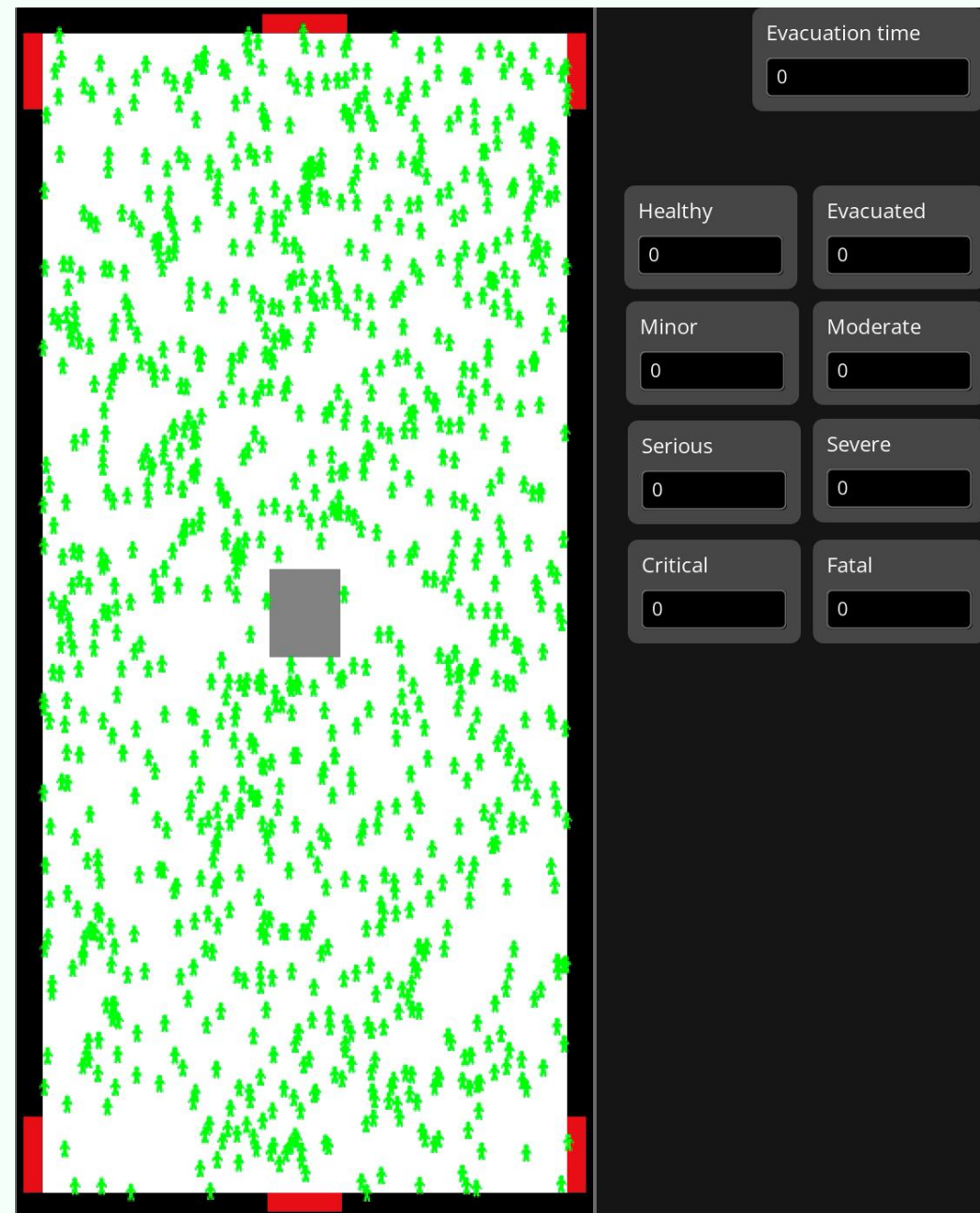
If the outcome is positive, the agent **slips**, **loses 5% of its current health** and does not move during that time step.



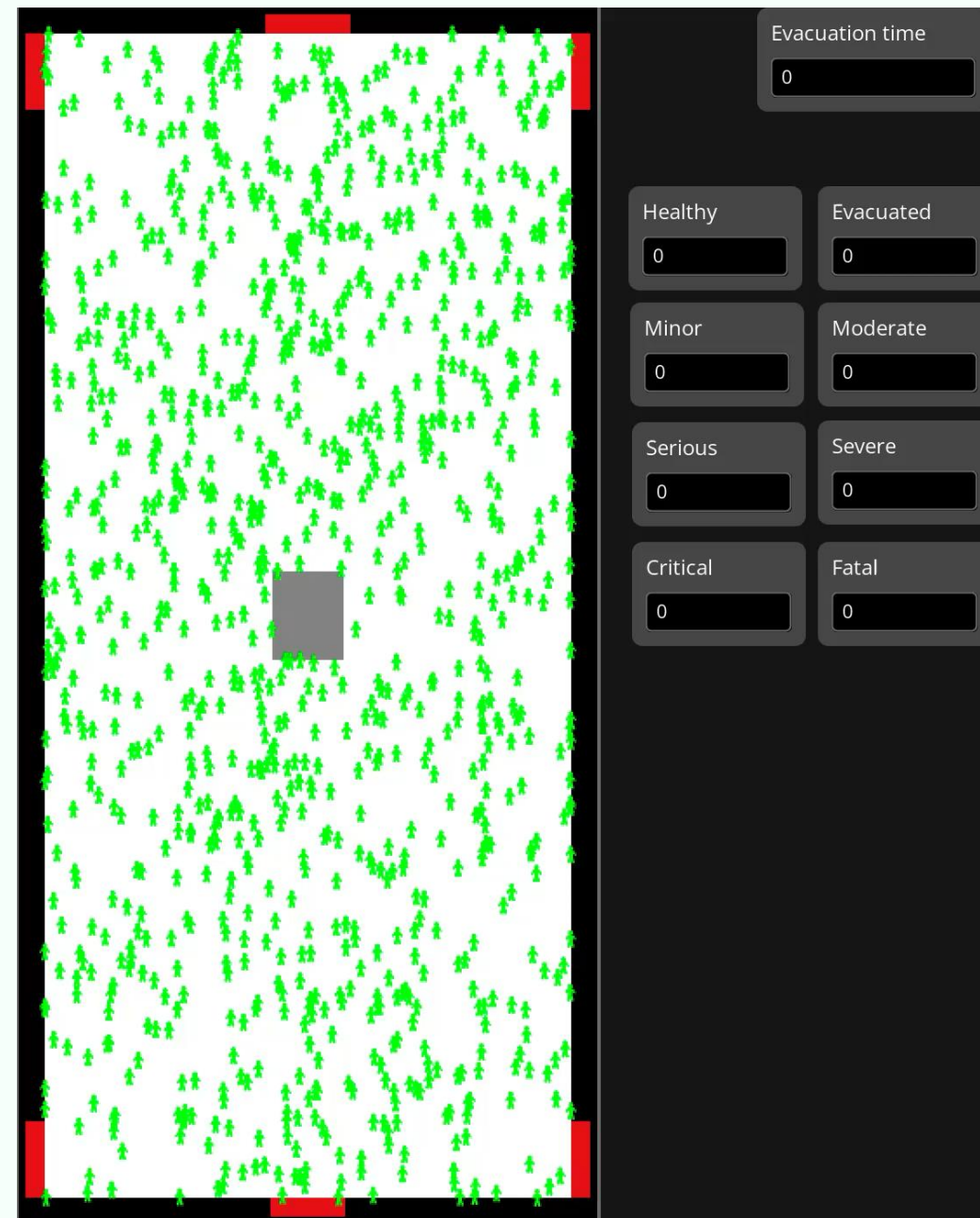
Sources of damage

Glass Bottles

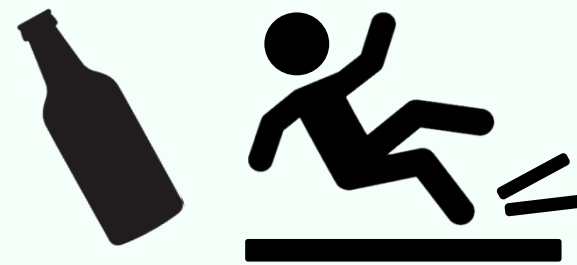
glass_bottles



glass_bottles



In the simulation, removing glass bottles was found to reduce massively the number of agents reaching the higher injury levels, making it one of the most **impactful safety measures** tested.



Sources of damage

Fire and smoke

The fire hazard can be activated via the *fire_hazard_enabled* switch.

fire_hazard_enabled

When enabled, a fire is introduced in the arena: after a delay controlled by *fire_delay* (number of ticks after the alarm), it ignites at a randomly selected patch and begins to spread.

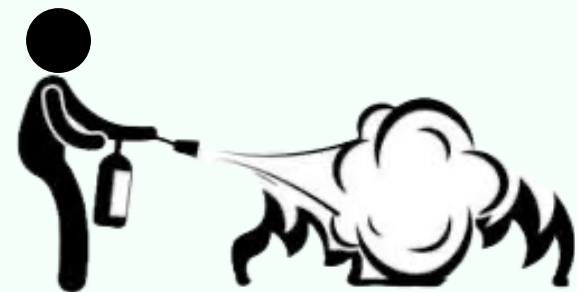
fire_delay

10

From that moment on, **the fire propagates stochastically at each tick** according to a *Probabilistic Cellular Automaton* model: every burning patch has a probability *fire_spread_rate* of igniting its four orthogonal neighbors, up to a maximum radius of 50 patches from the ignition point.

fire_spread_rate

0.2



Sources of damage

Fire and smoke



Agents standing directly on a burning patch lose health at a rate of `fire_damage_rate` per tick.

Agents on non-burning patches are not safe either: a **smoke diffusion system** propagates a field outward from the fire. The maximum possible smoke density is controlled by the `smoke_radius` parameter which defines the upper bound of the smoke scale.

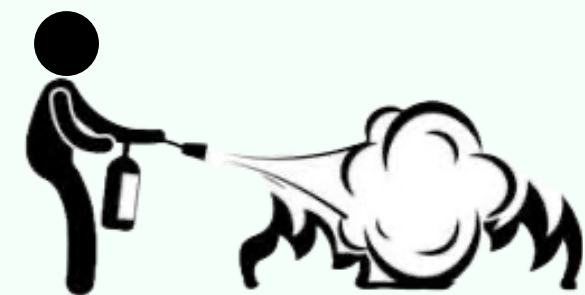
Agents caught in smoky areas suffer:

- **Inhalation damage**, ensuring that agents closer to the fire source take more damage than those at its fringes.

$$health_state_{t+1} = health_state_t - fire_damage_rate \cdot 0,3 \cdot \frac{smoke_level}{smoke_radius}$$

- **Disorientation**: when the local smoke level on an agent's patch reaches or exceeds the `smoke_visibility_threshold` the agent has a **40%** chance per tick of losing their current exit heading and being redirected to a random gate.

$$P(disorientation) = \begin{cases} 0,4 & \text{if } smoke_level > smoke_visibility_threshold \\ 0 & \text{otherwise} \end{cases}$$



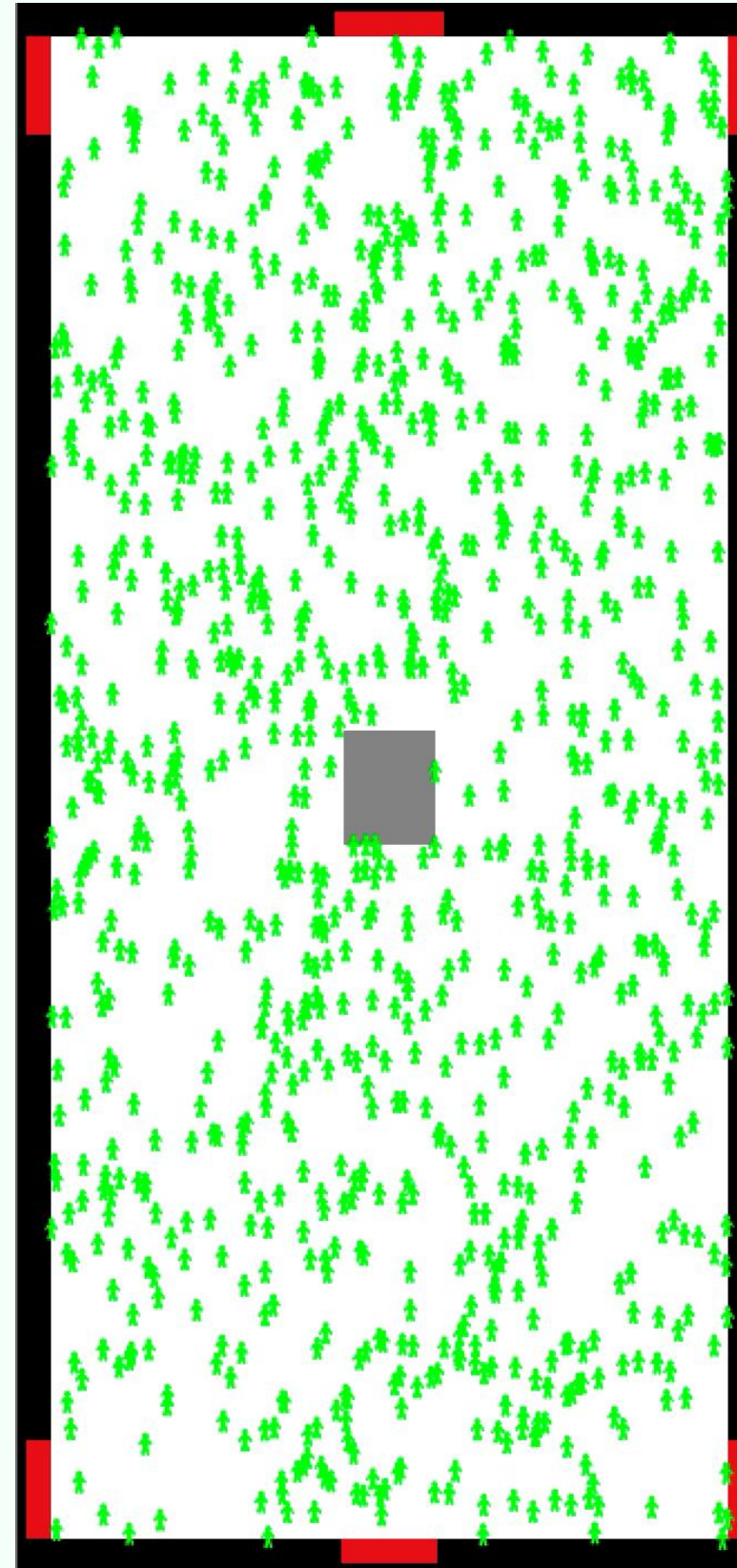
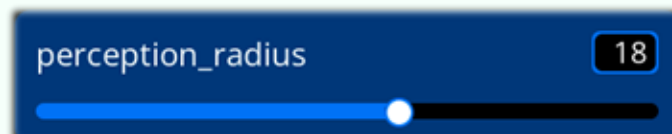
Sources of damage

Fire and smoke

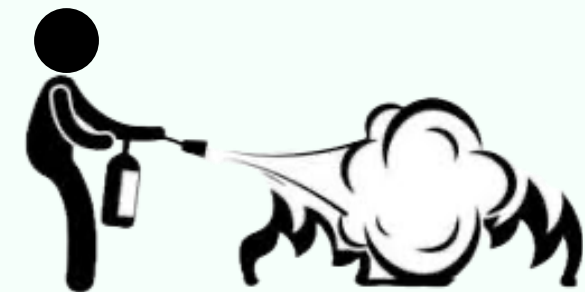
Aware agents

Can partially counteract all of this through the **deliberative BDI layer**, governed by:

The *perception_radius* defines **how far an aware agent can detect burning patches**, only fire within this radius enters the agent's belief state.



By setting `perception_radius` to its maximum value, the agents are able to maintain a safe distance from the fire for the entire duration of the simulation.



Sources of damage

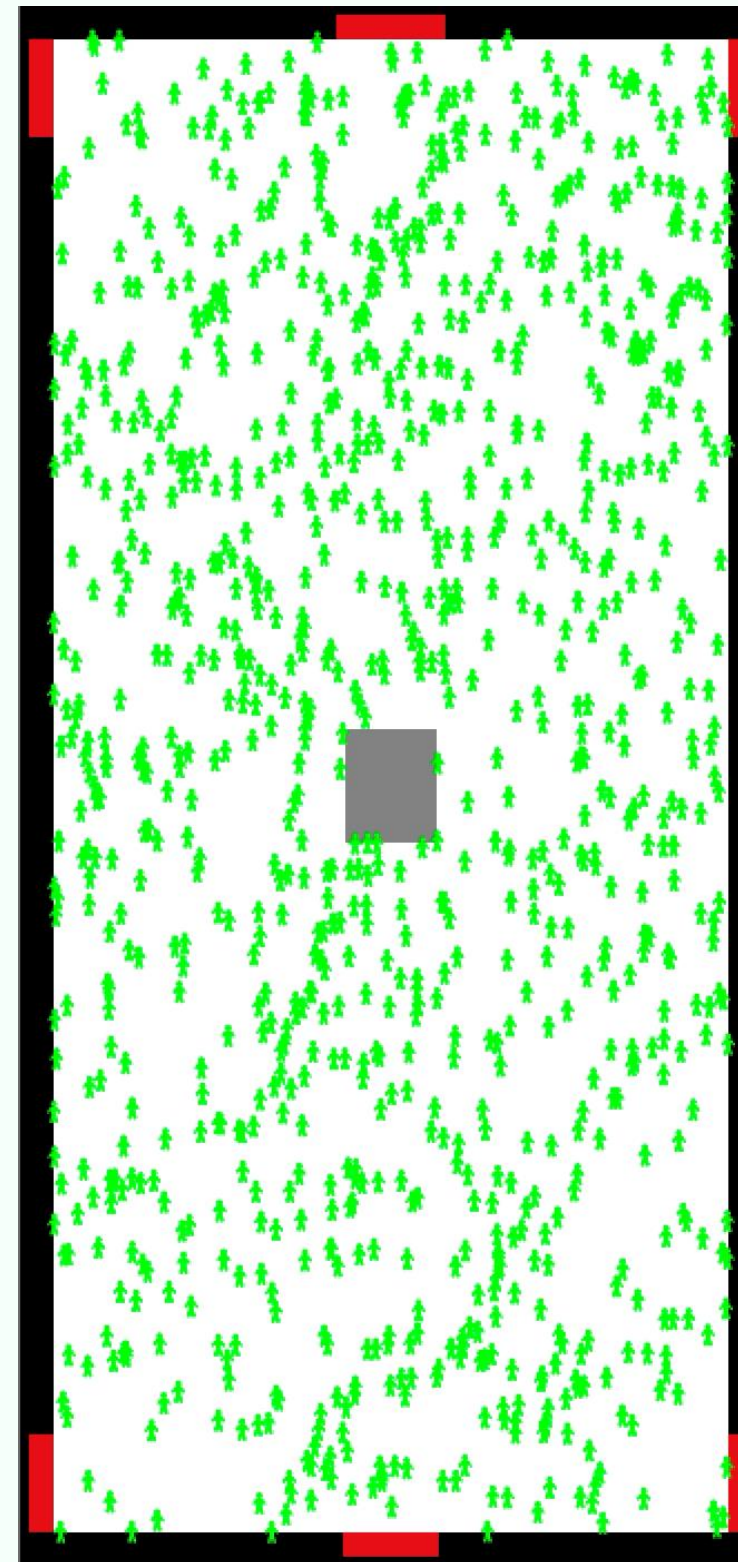
Fire and smoke

Aware agents

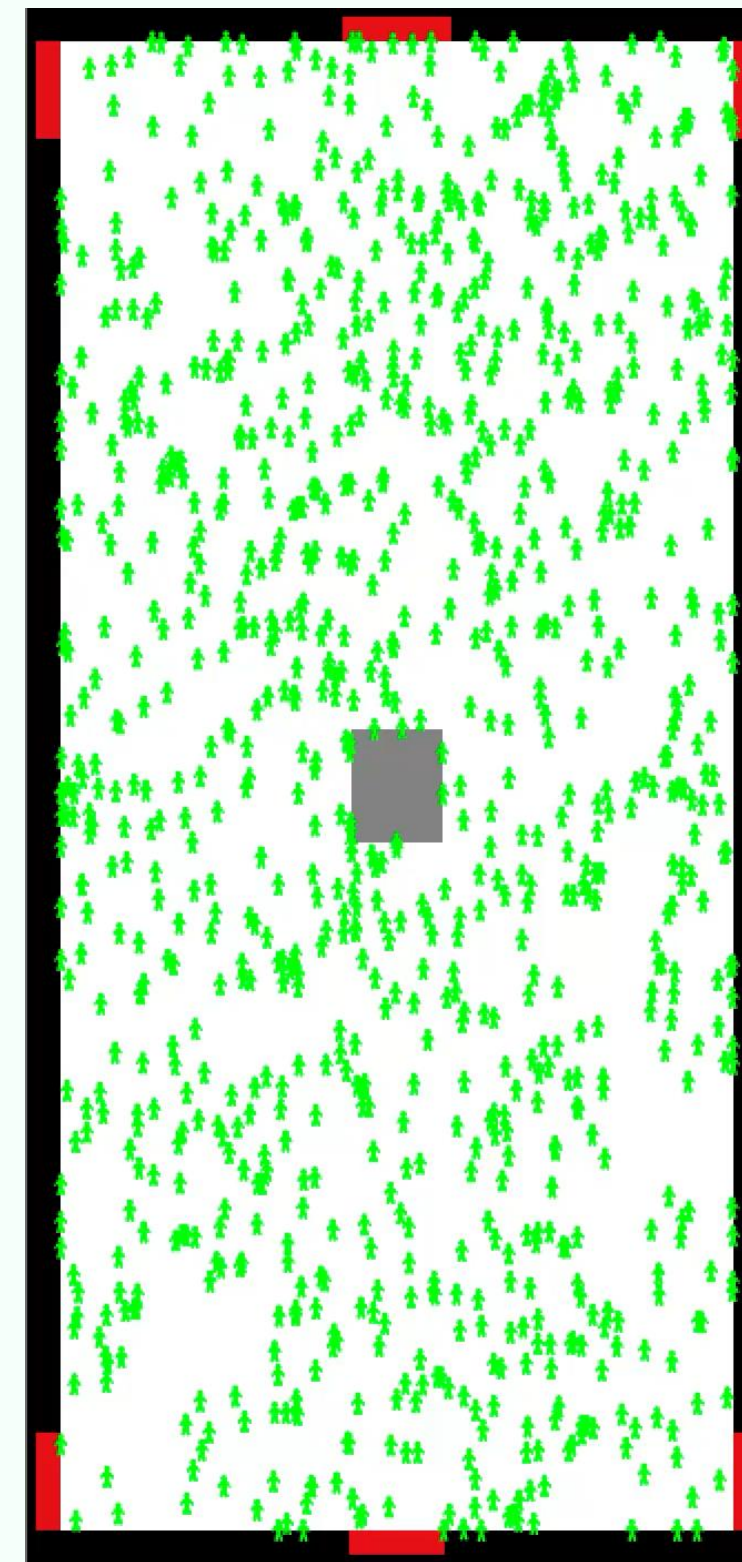
Can partially counteract all of this through the **deliberative BDI layer**, governed by:

The *safety_weight* controls **how strongly the agent prioritises fire avoidance** when re-evaluating exits every 5 ticks

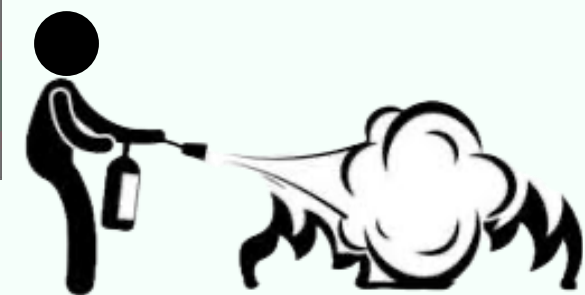
(a value of 0 means the agent ignores fire proximity entirely and simply heads to the nearest exit, while higher values make the agent willing to take a longer route in order to stay farther from the flames).



Safety_weight = 0

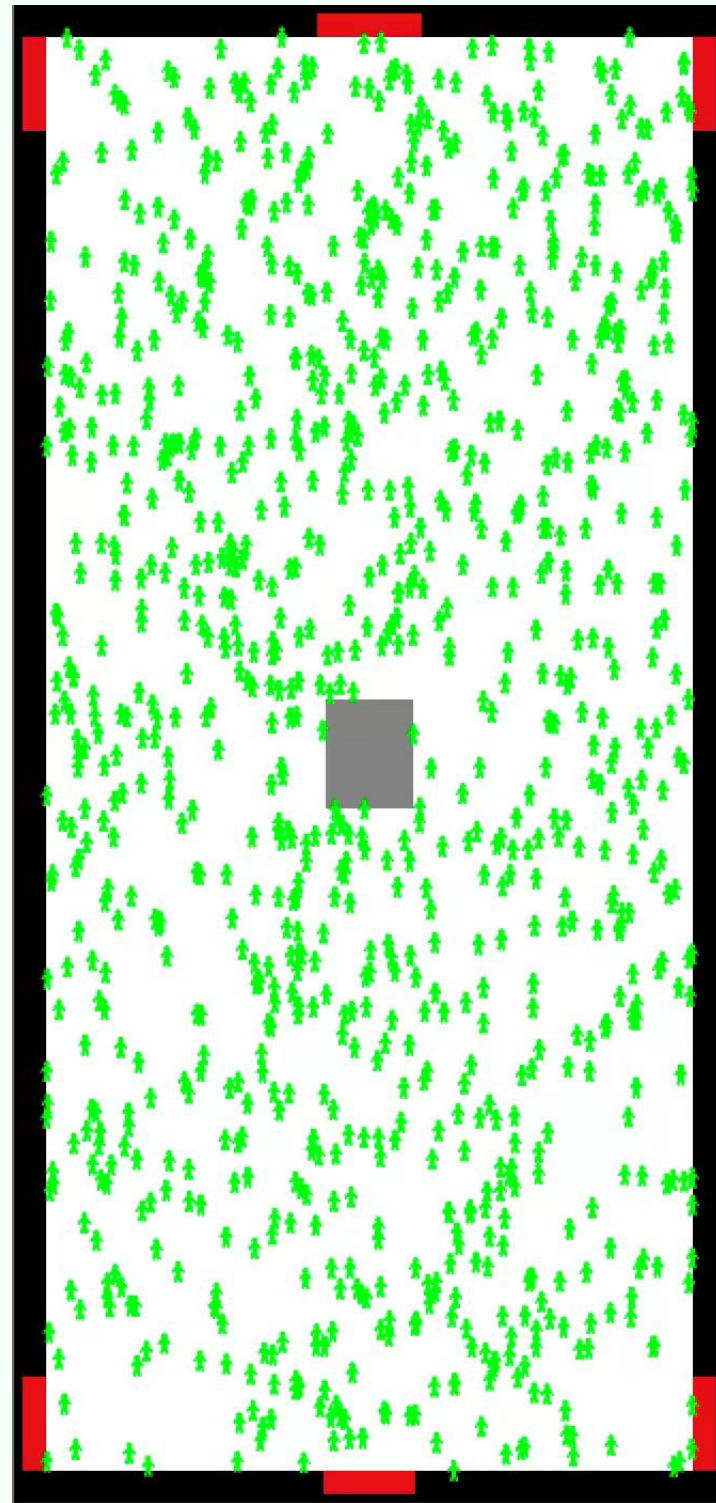


Safety_weight = 5

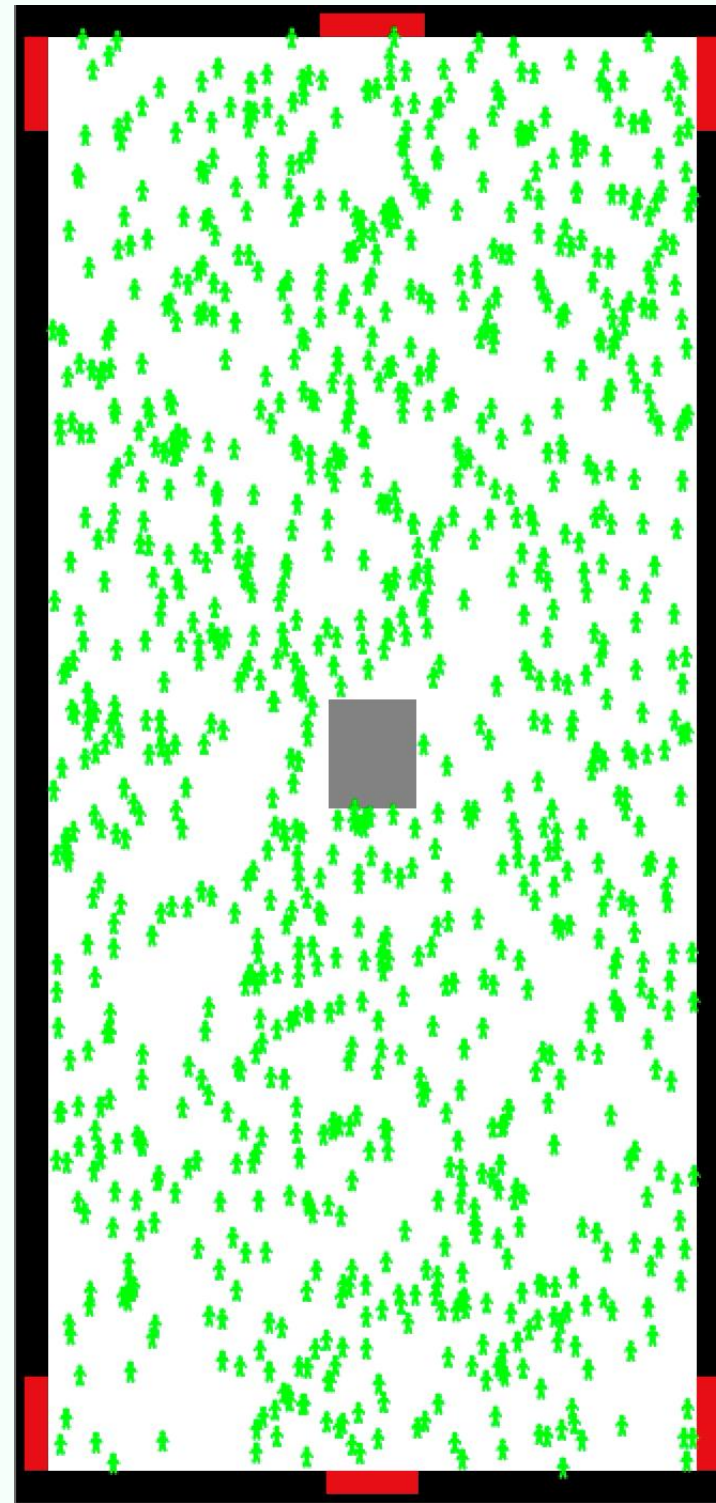


Sources of damage

Fire and smoke



Unaware agents

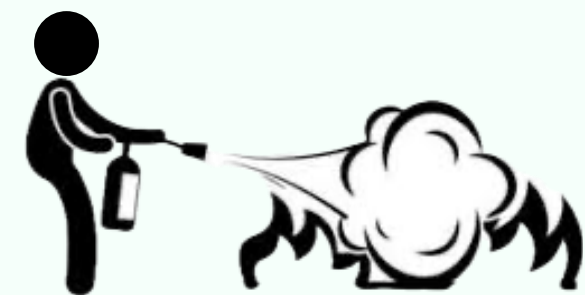


Unaware panicking agents

Unaware agents

Rely only on the **reactive layer** for fire avoidance, a pure stimulus-response mechanism that **diverts the agent away from a burning patch immediately ahead**, without any strategic replanning.

Panicking agents are particularly vulnerable. They behave normally on rational ticks, but during irrational ticks they rely on **follow_crowd**, a strategy with **no awareness of fire**, so they may blindly follow others onto burning patches, with no mechanism to avoid danger.

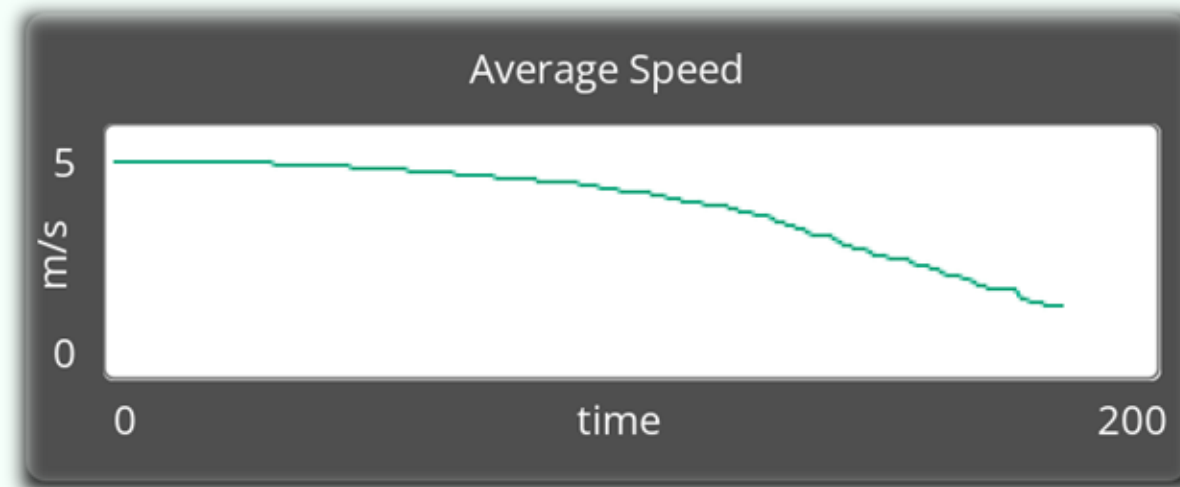


Plots

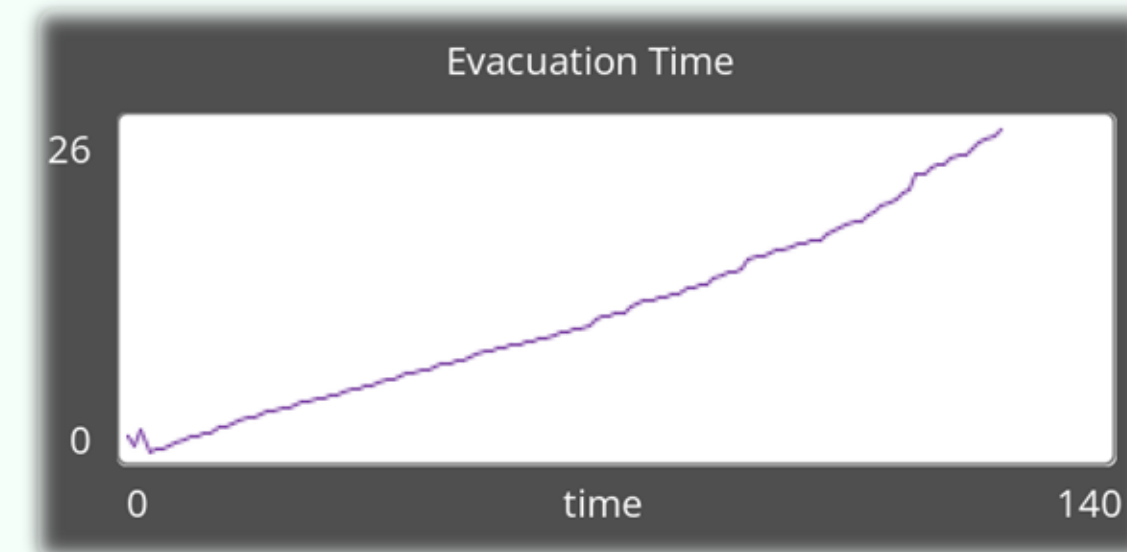
The **average speed plot** tracks the **mean movement speed of all alive agents at each tick**.

In normal conditions the curve is relatively stable, but it drops sharply as agents accumulate injuries and their speed is progressively reduced by the damage system.

The presence of panic, fire or dense smoke accelerates this decline.



The **evacuation time plot** tracks **how many agents have successfully exited the arena as a function of time**, giving a cumulative picture of the evacuation flow.

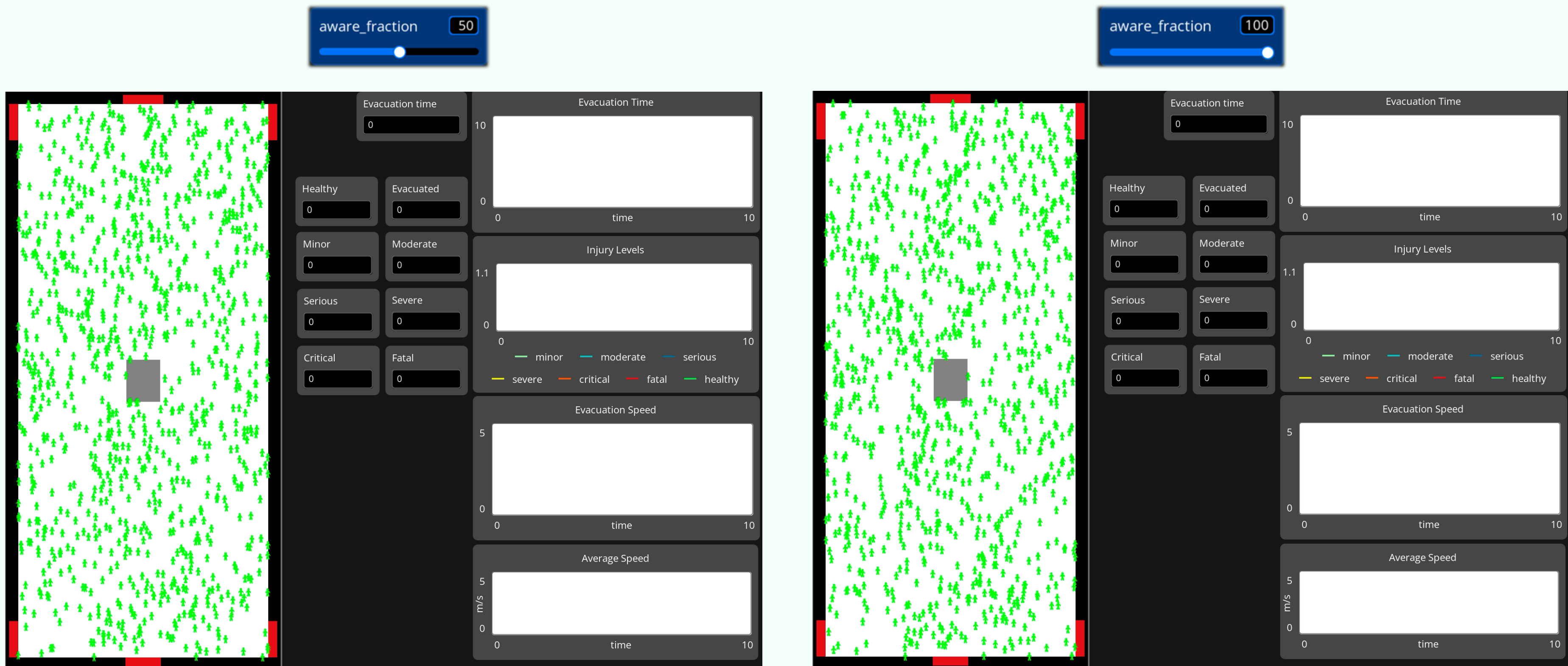


Experiments

Aware Agents



Let's suppose to have only **1000** adult male agents and let's analyse the `aware_fraction` in the simulation



Results

Aware Agents



Keeping the population fixed at **1000** male agents, increasing the aware fraction from 50% to 100% produces a **much faster evacuation process**: the total evacuation time is approximately halved.

This suggests that **increasing awareness greatly improves evacuation efficiency**, as more agents are able to move directly toward the nearest exit instead of dispersing toward suboptimal gates.

Experiments

Aware Agents



Now let's suppose to have **30.000** adult male agents and let's analyse the `aware_fraction` in the simulation



Results

Aware Agents



	Lower Awareness	Full Awareness
Evacuation Time	342	164
Total number of healthy agents	28453	25170
Total number of dead agents	3	26

These results show that **maximizing awareness is not always optimal** in very **dense crowds**: although a **fully aware crowd** reduces evacuation time, it also concentrates people near the same exits, **increasing congestion**, injuries, and fatalities.

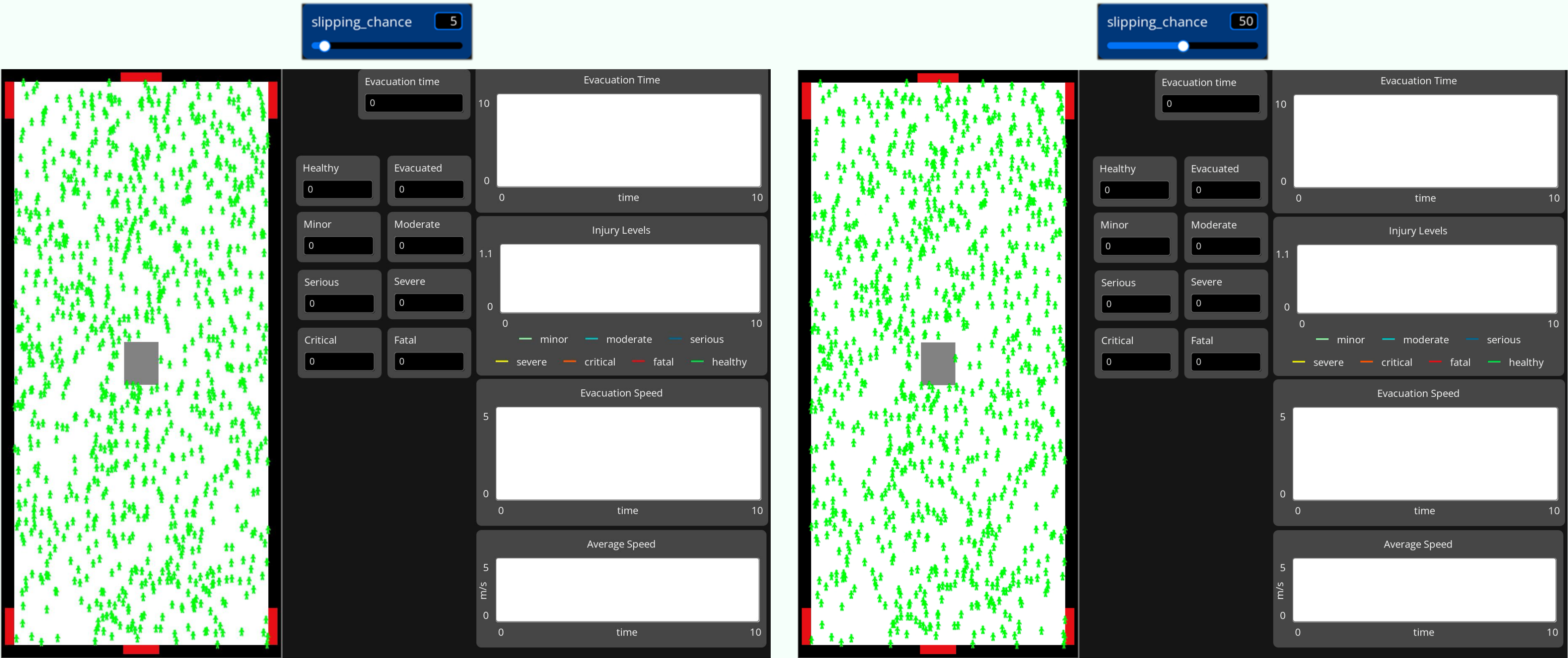
A **lower awareness** level slows the evacuation process, but leads to a **much lower death toll**.

Experiments

Glass Bottles



Let’s assume a population of 1000 adult male agents and let’s examine how the `slipping_chance` induced by glass bottles affects the simulation outcomes.



Results

Glass Bottles



Increasing the slipping chance dramatically **raises the level of danger**, leading to a mortality rate of about 80% of the population.

Conversely, substantially **reducing** the slipping chance still results in many **injuries**, but only a very small number of cases become fatal.

Conclusions



The simulation confirms that **faster** evacuation **does not always mean safer** evacuation.

Lower crowd density and safer environmental conditions reduce injuries and fatalities, whereas strategies that push too many agents toward the same exits can shorten evacuation time but increase mortality.

Therefore, effective **emergency planning** must **minimize** both **evacuation time** and **human losses**.



**Thanks for the
attention**

